Artificial Intelligence and Labor Market Dynamics: The Race between Human and AI[*]

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Abstract

This paper shows how modest aggregate labor market effects of artificial intelligence can coexist with substantial worker reallocation across occupations. I date the AI shock to the 2017 introduction of the Transformer architecture and, in an event-study difference-in-differences design, compare high- and low-AI-exposure occupations within skill groups. After 2017, the employment share of high-exposure occupations falls by 3.3 percentage points among middle-skill workers but rises by 3.0 percentage points among high-skill workers. Occupational switching, rather than unemployment, drives this divergence: middle-skill workers become more likely to leave high-exposure occupations and lose about 7 percent of earnings relative to comparable stayers, while high-skill workers move toward these occupations with no significant earnings change. Rising employer demand for AI skills in job postings and a steep education gradient in the switching response point to human-capital sorting. I formalize this mechanism in a frictional labor market model with endogenous AI growth.

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1 Introduction

Artificial intelligence is widely viewed as a general purpose technology. Eloundou et al. [2023] estimate that 80 percent of United States workers hold jobs in which large language models could affect at least 10 percent of tasks, and adoption spreads faster than any previous workplace technology [Bick et al., 2026]. Yet substantial disagreement remains about the labor market consequences of AI. Early studies find small aggregate effects [Acemoglu et al., 2022, Humlum and Vestergaard, 2026], while forecasts range from the displacement of half of entry-level white-collar jobs within five years [Amodei, 2025] to modest productivity effects over a decade [Acemoglu, 2025]. The disagreement partly reflects a gap in the evidence: Aggregate employment can conceal substantial worker reallocation across occupations. Which workers move, where they move, and what these moves cost them remain open questions. How do workers adjust in the early stages of the AI era, and what do these adjustments imply for the longer-run effects of AI?

I study these questions by comparing labor market outcomes across occupations that differ in both skill requirements and AI exposure. Skill requirements follow the O*NET Job Zone system, while AI exposure follows the suitability for machine learning measure of Brynjolfsson et al. [2018], validated by Eloundou et al. [2023]. I combine BLS occupation-level employment data with CPS worker transitions and wages. I date the onset of the recent AI wave to the 2017 introduction of the Transformer architecture [Mihet et al., 2025, Song et al., 2026] and compare high- and low-exposure occupations within skill groups. The central empirical result is a sharp divergence after the 2017 Transformer shock. The employment share of high-exposure occupations falls by 3.3 percentage points among middle-skill workers and rises by 3.0 percentage points among high-skill workers, and both changes depart sharply from pre-2017 trends. Worker-level event studies show that occupational switches, not unemployment, drive this divergence. Middle-skill workers become more likely to leave high-exposure occupations, while high-skill workers move toward them. The wage consequences are also asymmetric: middle-skill movers out of high-exposure occupations lose about 7 percent of annual earnings, while high-skill movers into them experience no significant earnings change.

To distinguish the mechanism behind these worker flows, I combine employer-side evidence from LinkUp job postings with worker-level heterogeneity in the

CPS. I use a large language model to code job descriptions and construct measures of both explicit AI requirements and the depth of AI use within each job. Employer demand for AI skills rises sharply after 2017: the share of postings with explicit AI requirements increases from about 0.2 percent in 2017 to 1.9 percent in 2025, while measures of AI intensity rise in parallel. On the worker side, the post-2017 increase in switching out of high-AI middle-skill occupations declines sharply with education: it is largest among workers with a high school education or less, smaller among those with some college, and approximately zero among workers with a bachelor's degree. Together, these patterns are consistent with a race between worker human capital and the AI frontier: workers who can keep pace remain in or move toward high-AI occupations, while workers who fall behind are more likely to switch toward lower-AI work.

To formalize the mechanism behind this reallocation, I develop a frictional labor market model with endogenous AI growth. Workers differ in human capital and search across High-AI and Low-AI jobs. High-AI employment advances the AI frontier through learning by using [Wang and Wong, 2026]. High-AI jobs operate closer to the frontier and allow workers to accumulate frontier-relevant skill more rapidly, but faster AI progress also raises the skill requirement for remaining in High-AI employment. This tension generates human-capital sorting: sufficiently skilled workers remain in or move toward High-AI jobs, while workers whose skills fail to keep pace reallocate toward Low-AI work. The model provides a framework that links the short-run worker flows in the data to the endogenous evolution of the AI frontier.

Related literature. This paper first contributes to the empirical literature on the labor market effects of AI. Existing evidence points to modest aggregate effects but larger consequences for particular workers and markets. Acemoglu et al. [2022] finds limited effects on broader employment and wages despite changes in vacancy requirements, while Humlum and Vestergaard [2026] documents rapid generative-AI adoption with modest early effects on employment and earnings. Other studies find concentrated losses among freelancers, young workers, junior positions, and workers in exposed occupations [Hui et al., 2024, Brynjolfsson et al., 2025, Hosseini Maasoum and Lichtinger, 2025, Song et al., 2026]. This paper helps reconcile these findings through worker reallocation across occupations. I use the 2017 introduction of the Transformer architecture as the AI shock and implement an event-study difference-in-differences design that compares high- and low-exposure occupations within skill groups. High-exposure occupations lose employment share among middle-skill workers but gain employment share among high-skill workers. These opposing responses can produce modest aggregate changes despite substantial reallocation across workers. Worker-level event studies show that occupational switches, not unemployment, account for the divergence. Middle-skill workers move away from high-exposure occupations and incur earnings losses, while high-skill workers move toward them without a significant earnings penalty. To examine why these flows move in opposite directions, I use LinkUp job postings to construct measures of realized employer demand for AI skills, rather than potential technological exposure alone [Brynjolfsson et al., 2018, Eloundou et al., 2023]. Measures of both explicit AI requirements and the depth of AI use rise sharply after 2017, which links worker reallocation to changes in the skills that employers demand from AI-intensive jobs.

The paper next contributes to models of technological change and labor market reallocation. The task-based framework of Acemoglu and Restrepo [2018] formalizes the race between automation and the creation of new tasks, and Acemoglu and Restrepo [2022] shows that task displacement accounts for a large share of the rise in U.S. wage inequality. Acemoglu [2025] applies the task-based framework to AI and predicts modest aggregate effects, while Freund and Mann [2026] studies how AI transforms jobs and induces specialization across tasks. This paper complements the task-based approach with a search-theoretic framework in which the incidence of AI operates through worker selection across occupations rather than through task displacement within occupations. Most closely related, Wang and Wong [2026] develops a search model in which AI adoption generates technological progress through learning by use and studies its implications for technological unemployment. This paper departs from that framework by making the effect of AI depend jointly on occupational exposure and worker human capital. Occupations differ in their exposure to AI, and workers differ in the human capital that determines access to AI-intensive work. High-AI jobs combine greater frontier productivity with higher skill requirements, and AI progress raises the qualification threshold for these jobs. The resulting selection mechanism draws sufficiently skilled workers toward AI-intensive occupations while pushing workers who fail to keep pace toward lower-AI work. Braxton and Taska [2023] provides a related mechanism in which technological change raises skill requirements and affects the consequences of job loss. The model places worker selection across heterogeneous occupations in a frictional labor market with endogenous AI growth and on-the-job search, and connects the observed worker flows to a race between worker human capital and the AI frontier.

More broadly, the paper relates to the literature on skill-biased technological change. A long tradition shows that technological progress raises relative demand for skilled labor and contributes to wage inequality [Katz and Murphy, 1992, Autor et al., 2003, Acemoglu and Autor, 2011]. This paper emphasizes occupational allocation as an additional margin of skill bias. AI does not affect workers with different skills uniformly within exposed occupations. Instead, technological progress sorts workers across occupations: middle-skill workers move away from high-AI work, while high-skill workers move toward it. The model further makes the strength of this sorting endogenous to technological progress because faster AI growth raises the skill requirement of AI-intensive jobs. This mechanism makes occupational allocation an endogenous margin through which technological progress can reshape the distribution of earnings.

The remainder of the paper proceeds as follows. Section 2 describes the data and the classification of occupations. Section 3 presents the empirical results. Section 4 presents the model. Section 5 concludes.

2 Data and Measurement

This section describes the data and the construction of the occupation groups used in the empirical analysis. Section 2.1 introduces the three main data sources. Section 2.2 defines occupational AI exposure. Section 2.3 constructs skill-by-exposure groups. Section 2.4 measures realized employer demand for AI skills from job postings. Section 2.5 introduces the timing of the AI shock.

2.1 Data

I use three complementary data sources to measure occupation-level employment, worker-level labor market transitions, and employer demand for AI skills.

Occupational Employment and Wage Statistics. I use the Bureau of Labor Statistics Occupational Employment and Wage Statistics (BLS OEWS) from 2012 to 2024. The AI exposure measure comes from Brynjolfsson et al. [2018], while occupational skill requirements come from O*NET Job Zones. These data use eight-digit O*NET-SOC codes, whereas BLS reports employment and wage statistics at the six-digit SOC level. I therefore map occupations to their corresponding six-digit SOC codes and merge the data at that level. I drop occupations with missing employment in any year from 2012 to 2024. The final balanced sample contains 656 occupations observed over 13 years and covers approximately 104 million workers in 2012 and 119 million workers in 2024.

Current Population Survey. I use the Current Population Survey (CPS) monthly files to measure worker-level labor market transitions and the Annual Social and Economic Supplement (ASEC) to measure annual earnings. The baseline sample consists of civilian workers of both sexes aged 18 to 60 who are employed at the time of the survey and report a valid occupation; I drop military occupations, observations with missing or invalid occupation codes, and observations with invalid usual weekly hours. The rotating panel structure of the CPS allows me to link individuals across survey months using consistent person identifiers, so I can distinguish occupational switches from transitions into unemployment or out of the labor force. All statistics are weighted using the CPS final person weights. CPS occupations are recorded in Census occupation codes, which I map to six-digit SOC codes before assigning workers to the occupation groups defined below. Because the CPS switches from the 2010 to the 2018 Census occupation classification in January 2020, I apply separate Census-to-SOC crosswalks for the two coding regimes. Workers who match to one of the 656 occupations in the baseline sample (633 of which appear in the CPS after the crosswalk) are retained for analysis. The resulting sample contains approximately 4.7 million individual-month observations over 2017–2025.

Sample Coverage and Consistency for BLS and CPS. The final BLS-OEWS sample contains 656 six-digit SOC occupations observed from 2012 to 2024. After the Census-to-SOC crosswalk, the CPS covers 633 occupations and contains approximately 4.7 million individual-month observations over 2017–2025. The occupation coverage is similar across the two datasets, and the harmonized CPS sample closely reproduces the employment patterns in BLS OEWS. Between 2017 and 2024, the employment share of middle-skill high-exposure occupations falls by 3.8 percentage points in the CPS and 3.3 percentage points in BLS OEWS, while the share of high-skill high-exposure occupations rises by 3.5 and 3.0 percentage points, respectively. Appendix Tables A.1 and A.2 report the corresponding occupation counts, employment levels, and employment shares.

LinkUp Job Postings. To measure labor demand and job requirements at the vacancy level, I use the universe of U.S. online job postings from LinkUp. LinkUp is a commercial job market data provider that has indexed job postings since 2012 by scraping employer websites directly, rather than aggregating listings from job boards. This sourcing has two advantages. First, because every posting is collected from the hiring company's own website, the data largely avoid the duplicate, expired, and third-party listings that contaminate job-board aggregators. Second, LinkUp revisits each website daily and records both the date a posting first appears and the date it is taken down, so the data capture the full life cycle of a vacancy rather than a single snapshot. The full sample contains 264.8 million U.S. postings from January 2012 through September 2025. For each posting, I observe the job title, employer, location, posting and removal dates, and the full text of the job description; LinkUp also assigns each posting an eight-digit O*NET-SOC occupation code based on its title and content. I truncate this code to the six-digit SOC level and match each posting to the same five occupation groups used throughout the paper, which allows me to track vacancy postings and their stated skill requirements for exactly the same occupation groups that I follow in the BLS and CPS data.

Representativeness of LinkUp. Because LinkUp collects postings from employer websites rather than from a probability sample, I validate its representativeness against the CPS. Figure A.7 shows the results. LinkUp vacancy shares closely match CPS employment shares in both levels and 2017–2024 changes for Groups 2 through 5. The only notable gap is Group 1, whose minimal-preparation jobs are less likely to be advertised on employer websites and are therefore underrepresented in postings. Since Group 1 plays no role in the posting-level analysis, including or excluding it leaves the main results unchanged.

The three sources capture different margins of labor market adjustment. BLS measures changes in employment across occupations, CPS identifies the worker flows behind these changes and their earnings consequences, and LinkUp measures changes in employer demand.

2.2 Occupational AI Exposure

The baseline measure of AI exposure is the suitability for machine learning (SML) score of Brynjolfsson et al. [2018]. Their rubric evaluates work activities along 23 dimensions of machine-learning suitability, covering 18,156 detailed work activities, and aggregates these ratings to the occupation level using O*NET task-importance weights. I use their published occupation-level score, denoted mSML, which captures the technological exposure of an occupation to machine learning. I also validate the measure against the large-language-model exposure measure of Eloundou et al. [2023]. The two measures display a strong positive relationship across occupations shown in Figure A.1.

Within each skill group, I divide occupations at the median of the SML distribution. Occupations above the within-group median constitute the high-exposure group, while occupations below the median constitute the low-exposure group. I also use exposure quintiles within each skill group to examine whether labor market responses vary systematically with AI exposure.

2.3 Skill Requirements and Occupation Groups

I measure occupational skill requirements with the O*NET Job Zone classification. Job Zones summarize the education, experience, and preparation that an occupation typically requires. I classify Job Zones 4 and 5 as high-skill occupations and Job Zones 2 and 3 as middle-skill occupations, while Job Zone 1 forms a residual low-skill group. This ranking is closely aligned with the conventional distinction between low-, middle-, and high-skill occupations in the labor economics literature [Acemoglu and Autor, 2011, Autor and Dorn, 2013, Cortes, 2016].

I further validate the O*NET-based classification using workers' observed educational attainment in the CPS. Workers in high-skill occupations are predominantly college graduates, while workers in middle-skill occupations are more likely to hold a high school diploma or some college education. Appendix Figure A.2 documents the close correspondence between the O*NET classification and workers' observed education. Educational composition is also similar between high- and low-AI-exposure occupations within each skill tier, supporting comparisons across exposure groups within the same skill category.

The intersection of skill requirements and AI exposure therefore defines five occupation groups: Group 1 (low skill), Group 2 (middle skill, low AI), Group 3 (middle skill, high AI), Group 4 (high skill, low AI), and Group 5 (high skill, high AI). The main analysis focuses on Groups 2–5, which form a two-by-two design in skill requirements and AI exposure. Group 1 serves as a separate residual group.[1] Appendix Table A.1 reports the group definitions, representative occupations, and occupation coverage in the BLS-O*NET and CPS samples.

2.4 Realized Employer Demand for AI Skills

Occupation-level exposure measures capture the technological potential for AI to affect an occupation [Brynjolfsson et al., 2018, Eloundou et al., 2023], but they do not reveal whether employers actually demand AI-related skills. I therefore use the full text of LinkUp job postings to construct measures of realized employer demand for AI skills.

I focus on the two high AI-exposure groups, Group 3 and Group 5, and draw a reproducible stratified random sample of up to 5,000 U.S. postings per group and year from 2017 onward. The sample contains 72,759 postings, of which 97.4 percent have a nonempty job description. This restriction leaves 70,832 postings for the text analysis.

I code each posting independently with GPT-4o-mini. The model reads the job description and records education and experience requirements, task complexity, and AI skill content. It also records a short excerpt from the original posting that supports each classification, which permits direct verification of the coded attributes.

I construct two measures of AI skill demand. The first, *Explicit AI Requirement*, equals one if the posting explicitly requires AI- or machine-learning-related skills,

[1]Employment in Job Zone 1 occupations remains nearly flat over the sample period, with no visible break after 2017 (Appendix Figure A.4). Adding them or not does not influence my main analysis. I therefore exclude Group 1 from the main two-by-two analysis.

such as machine learning, natural language processing, or TensorFlow. This measure captures the extensive margin of AI skill demand. The second, *AI-Use Intensity*, ranges from 1 to 5 and measures how central AI is to the job. A score of 1 indicates no meaningful AI use, while a score of 5 indicates responsibility for the design or strategic direction of AI systems. Intermediate values capture increasingly intensive use of AI tools, models, and workflows. I classify a posting as AI intensive when its score is at least three.

As a robustness check, I construct an independent keyword-based measure from a dictionary of 31 AI-related terms. The dictionary draws on canonical references in the AI literature [LeCun et al., 2015, Zhang et al., 2022] and topic classifications from major AI journals and conferences, including IEEE and ACM. It covers terms such as machine learning, deep learning, computer vision, natural language processing, and robotics.

Section 3.3 reports the evolution of these measures over time.

2.5 Timing of the AI Shock: The 2017 Transformer

I date the AI shock to June 2017, when Vaswani et al. [2017] introduces the Transformer architecture. Transformer-based architectures revolutionize the ability of AI systems to process and learn from unstructured data and underpin modern foundation models such as BERT and GPT. Their introduction therefore represents a discrete shift in AI capability and marks the onset of the recent AI wave. Consistent with this interpretation, recent economics studies also use 2017 as the timing of the AI shock [Mihet et al., 2025, Song et al., 2026].

Patent data from the United States Patent and Trademark Office also provide independent support for 2017 as a technological break. Figure A.3 compares patent growth across 498 Cooperative Patent Classification subclasses. The AI and machine-learning subclass, G06N, records the fastest post-2017 growth and shows a clear acceleration after 2017. This pattern is consistent with the introduction of the Transformer architecture marking a discrete shift in the trajectory of AI innovation rather than an arbitrary sample split.

The event-study specifications thus use 2016 as the omitted reference year. The coefficients therefore measure changes in the difference between high- and low-exposure occupations relative to the corresponding difference immediately before the AI shock. Pre-2017 coefficients provide a direct assessment of differential pre-trends.

3 Empirical Results

The empirical analysis proceeds in three steps. First, I document opposite post-2017 employment responses to AI exposure across skill groups. Second, I show that occupational switching, rather than unemployment, drives this divergence, with substantial earnings losses concentrated among middle-skill workers who move away from high-AI occupations. Third, I provide evidence that these opposing flows reflect human-capital sorting as demand for AI-related skills rises.

3.1 Fact 1: Employment shares diverge across skill groups within exposure levels

Using BLS OEWS data, I first examine how employment shares evolve around the 2017 AI shock. Figure 1 plots changes in employment shares relative to 2017, $s_t - s_{2017}$, together with linear pre-2017 trends. The gray series report the corresponding aggregate patterns when occupations are pooled across skill groups.

At the aggregate level, high- and low-AI-exposure occupations show relatively small differences in employment dynamics after 2017. This pattern mirrors the modest aggregate labor market effects documented in earlier work [Acemoglu et al., 2022, Humlum and Vestergaard, 2026]. The aggregate series, however, conceal sharply different responses across skill groups.

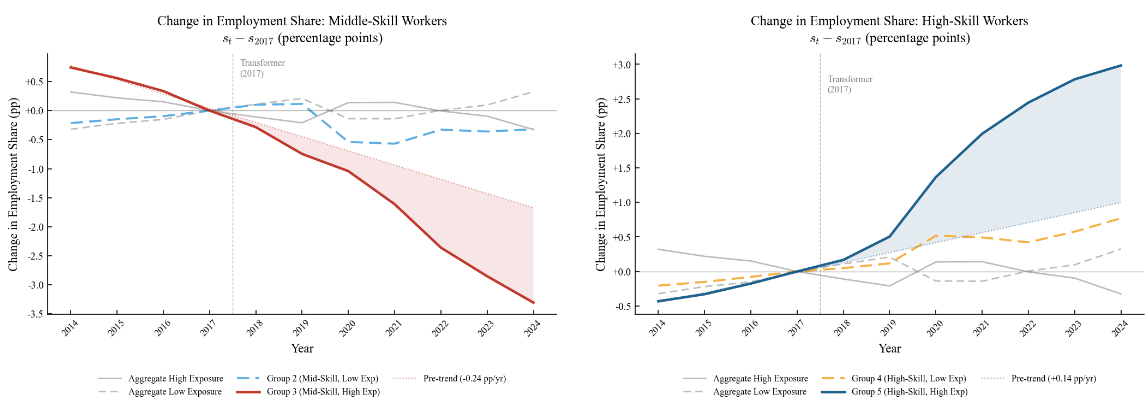
Among middle-skill occupations, the employment share of the high-exposure group, Group 3, falls by 3.3 percentage points between 2017 and 2024. Its pre-2017 trend is only -0.24 percentage points per year, and the post-2017 path moves progressively below the extrapolated trend. Employment in the middle-skill low-exposure group, Group 2, remains approximately flat.

The pattern reverses among high-skill occupations. The employment share of the high-exposure group, Group 5, rises by 3.0 percentage points between 2017 and 2024, relative to a pre-2017 trend of only $+0.14$ percentage points per year. Group 4, the high-skill low-exposure group, increases by less than one percentage point.

The opposing responses explain how large worker reallocation can coexist with small aggregate employment effects. High-exposure occupations lose employment share among middle-skill workers but gain almost the same amount among high-skill workers. Aggregation across skill groups therefore offsets the two responses and masks the underlying reallocation.

Moreover, the skill-specific divergence becomes stronger with AI exposure. Figure 2 replaces the binary exposure measure with SML quintiles within each skill group. Among middle-skill occupations, the decline is concentrated in the highest-exposure quintile, which falls by roughly 2.6 percentage points, while the remaining quintiles exhibit much smaller changes. Among high-skill occupations, the largest gains occur in the upper part of the exposure distribution, particularly the top two quintiles, while changes are more modest at lower exposure levels. This pattern therefore links the post-2017 reallocation to AI exposure itself rather than to a correlated occupation characteristic.

The same reallocation appears in CPS household data (Appendix Table A.2), and LinkUp postings show the matching shift in labor demand (Appendix Figure A.8).

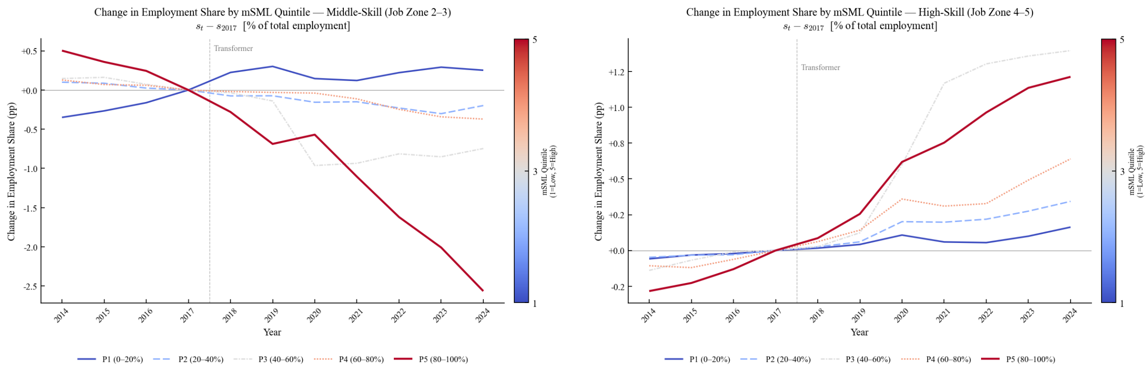


***(a)** Middle-skill occupations*

***(b)** High-skill occupations*

***Figure 1:** Aggregate Stability and Skill-Specific Employment Reallocation*

Notes: The figure plots changes in employment shares relative to 2017, $s_t - s_{2017}$, together with linear pre-2017 trends. The gray series show the corresponding aggregate changes for high- and low-AI-exposure occupations when skill groups are pooled. Group 3 (middle skill, high AI) falls by 3.3 percentage points between 2017 and 2024, while Group 5 (high skill, high AI) rises by 3.0 percentage points. Source: BLS OEWS.



(a) *Middle-skill occupations* ***(b)*** *High-skill occupations*

Figure 2: *Employment Reallocation across the AI-Exposure Distribution*

Notes: Each panel plots the change in employment share relative to 2017 by quintile of the occupation-level SML measure within each skill group. Among middle-skill occupations, the decline is concentrated in the highest AI-exposure quintile. Among high-skill occupations, employment gains are largest in the highest-exposure quintile. Source: BLS OEWS.

3.2 Fact 2: Occupational mobility, not unemployment, drives reallocation

Fact 1 shows that employment shares diverge sharply across skill groups after the 2017 Transformer shock. I next examine the worker-level adjustment around this technological break. I estimate event-study specifications separately for middle- and high-skill workers that compare transition outcomes between high- and low-AI-exposure occupations before and after 2017:

$$Y_{it}^{j} = \alpha_j + \sum_{k \neq 2017} \beta_k^j \left(\text{Year}_{t=k} \times \text{HighAI}_i\right) + \Gamma_j X_{it} + \delta_m + \epsilon_{it}, \tag{1}$$

where $j \in \{M, H\}$ denotes the skill group. The outcome Y_{it}^j equals, in turn, an indicator for exit from the initial occupation group, unemployment, unemployment or nonparticipation, or an occupational switch. The variable HighAI_i equals one for workers in high-AI-exposure occupations. The vector X_{it} includes a cubic polynomial in age, sex, race, nativity, education, and marital status, and δ_m denotes month fixed effects. I normalize 2017, the year of the Transformer shock, to zero. The coefficient β_k^j therefore measures the high- versus low-exposure difference in year k relative to the corresponding difference at the time of the shock.

Figure 3 reports the estimates for middle-skill workers. Before the Transformer shock, the coefficients are small and close to zero across all four outcomes. After 2017, workers in high-exposure occupations become increasingly more likely to leave their initial occupation group than workers in low-exposure occupations. The differential exit probability reaches approximately 0.5 to 0.75 percentage points per month by 2022 and 2023. Occupational switches account for most of this increase. In contrast, transitions into unemployment or nonparticipation show little persistent differential response outside the pandemic period.

Figure 4 shows the opposite response among high-skill workers. Before 2017, high- and low-exposure occupations display similar transition patterns. After the Transformer shock, workers in high-exposure occupations become less likely to exit or switch than workers in low-exposure occupations. Equivalently, high-skill workers in low-exposure occupations become relatively more likely to move. Transitions into unemployment again show little persistent differential response.

Moreover, the pooled estimates mask substantial heterogeneity by age. Appendix Figure A.9 shows that the post-2017 increase in occupational switching among exposed middle-skill workers is concentrated among older workers. Among high-skill workers, in contrast, the relative movement toward high-AI occupations is stronger among younger workers. The age pattern is consistent with a life-cycle interpretation: older middle-skill workers are more likely to move away from AI-exposed work, while younger high-skill workers are better positioned to move toward occupations that benefit from the new technology.

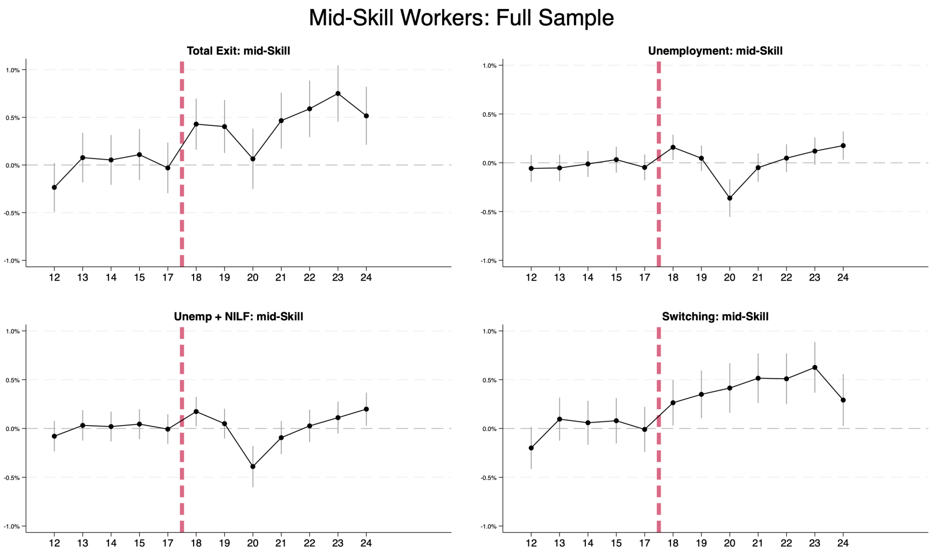


Figure 3: *Event Study for Middle-Skill Workers: High-Exposure versus Low-Exposure*

Notes: The figure plots β_k^M from equation (1) with 95 percent confidence intervals, for four outcomes: total exit, unemployment, unemployment plus nonparticipation, and occupational switch. Coefficients are flat before 2017 for all outcomes. After 2017, total exit and occupational switches rise for high-exposure workers, while transitions into unemployment and into unemployment or nonparticipation show no differential trend outside the pandemic year 2020. Source: CPS.

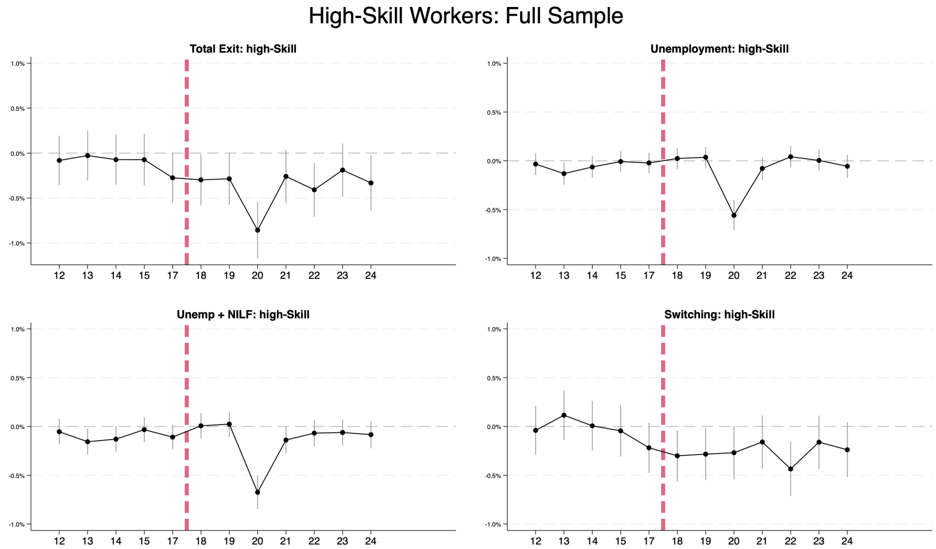
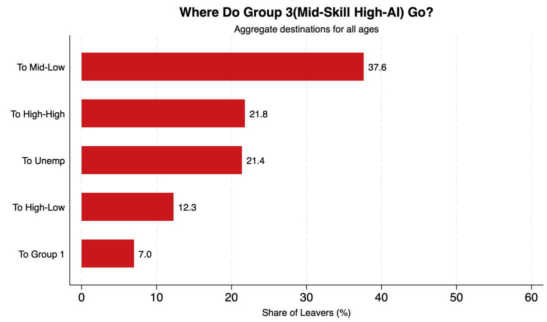


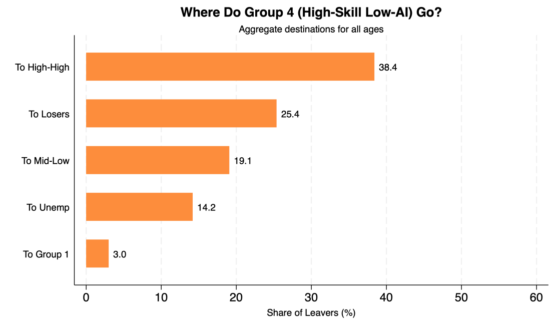
Figure 4: *Event Study for High-Skill Workers: High-Exposure versus Low-Exposure*

Notes: The figure plots β_k^H from equation (1) with 95 percent confidence intervals. After 2017, high-exposure workers exhibit lower total exit and lower switch rates than low-exposure workers, which implies that low-exposure high-skill workers switch more. Source: CPS.

Destinations of movers. Destination shares clarify the direction of occupational reallocation. Figure 5 reports the destinations of workers who leave Group 3 (middle skill, high AI) and Group 4 (high skill, low AI). Among Group 3 leavers, Group 2 (middle skill, low AI) is the most common destination, accounting for 37.6 percent of moves. Among Group 4 leavers, Group 5 (high skill, high AI) is the most common destination, accounting for 38.4 percent. Thus, conditional on leaving their initial occupation group, middle-skill workers disproportionately move toward lower-AI-exposure work, while high-skill workers disproportionately move toward higher-AI-exposure work. These destination patterns mirror the employment-share changes in Fact 1: the high-exposure share falls within the middle-skill tier but rises within the high-skill tier. Appendix Table A.3 reports the corresponding changes in employment shares by education group in the CPS.



(a) Destinations for Group 3 movers



(b) Destinations for Group 4 movers

Figure 5: *Destinations of Movers: Within-Skill Reallocation*

Notes: Each panel reports the share of leavers by destination, aggregated over all ages. Middle-skill high-exposure workers move mainly to middle-skill low-exposure occupations. High-skill low-exposure workers move mainly to high-skill high-exposure occupations. Source: CPS.

Earnings consequences. The destination evidence shows that middle- and high-skill workers move in opposite directions across AI-exposure groups. I next examine whether these moves carry different earnings consequences. I compare annual earnings changes between movers and stayers within each skill group:

$$\Delta W_{i,j,t} = \alpha + \beta \cdot \mathbb{1}(\text{Mover}_{i,t}) + \mathbf{X}'_{i,t}\gamma + \delta_{jt} + \epsilon_{i,j,t}, \tag{2}$$

where $\Delta W_{i,j,t}$ is the change in real annual earnings for individual i in industry j and year t. The vector $\mathbf{X}_{i,t}$ includes a cubic polynomial in age and controls for sex, race, nativity, education, and marital status, while δ_{jt} denotes industry-by-year fixed effects. The baseline estimates use inverse probability weights based on pre-move earnings and worker characteristics.

Table 1 reports sharply different earnings consequences across the two directions of reallocation. Middle-skill workers who move from high- to low-AI-exposure occupations lose \$1,971 in annual earnings relative to comparable stayers, or 7.15 percent of their pre-move earnings. In contrast, high-skill workers who move from low- to high-AI-exposure occupations experience a statistically insignificant earnings change of $-$\$696, or 1.20 percent of pre-move earnings.

The earnings asymmetry gives economic content to the worker flows. Middle-skill workers move away from AI-exposed occupations at a substantial earnings cost, while high-skill workers move toward AI-exposed occupations without a detectable earnings penalty. Together with the transition evidence, these results show that the post-2017 adjustment operates through occupational reallocation rather than persistent unemployment, with the adjustment costs concentrated among middle-skill workers.

Table 1: *IPW-Adjusted Wage Dynamics: Movers versus Stayers*

	Coefficient $\hat{\beta}$ (\$1000)	**Pre-Move Wage** (\$1000)	**Magnitude** (% of Wage)
Panel A: Middle skill group			
High $\rightarrow$ Low AI	-1.971^{**}	27.58	-7.15%
	(0.881)		
$N = 23{,}858$	$R^2 = 0.073$		
Panel B: High skill group			
Low $\rightarrow$ High AI	-0.696	58.02	-1.20%
	(1.365)		
$N = 14{,}781$	$R^2 = 0.059$		

Notes: The table reports estimates of β from equation (2) with inverse probability weights. The dependent variable is the change in real annual wages in thousands of dollars. IPW reweights are based on pre-move wage, age, sex, education, marital status, nativity, and race, with propensity scores trimmed at the 2nd and 98th percentiles. Controls include cubic age polynomials, demographics, and industry-by-year fixed effects. Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: CPS.

3.3 Fact 3: Rising AI skill requirements generate human-capital sorting

The opposing worker flows in Facts 1 and 2 raise a natural question: why do middle- and high-skill workers respond differently within AI-exposed occupations? I use LinkUp job postings and worker-level heterogeneity in the CPS to examine a human-capital mechanism.

Figure 6 shows that AI-specific skills become increasingly important in job postings after the 2017 Transformer shock. The share of postings with explicit AI requirements rises from about 0.2 percent in 2017 to about 1.9 percent in 2025, while the share with AI-use intensity of at least three rises from about 0.2 percent to about 1.3 percent. Both measures increase roughly six- to eightfold over the sample period. The increase is highly concentrated in high-skill exposed occupations. In

Group 5, the share of postings with explicit AI requirements rises from 0.46 percent in 2017 to 2.29 percent in 2024 and 3.80 percent in 2025, while the corresponding share in Group 3 remains below 0.4 percent throughout the sample.

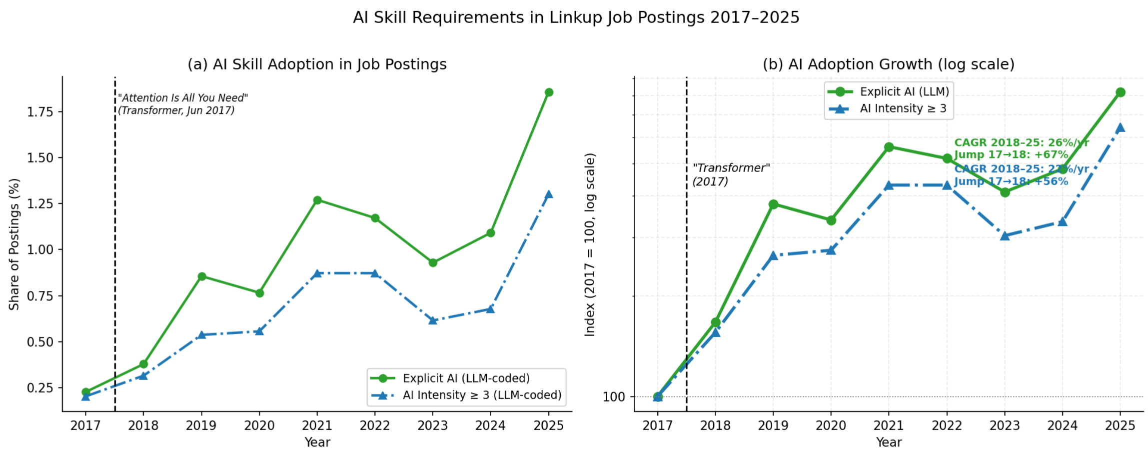


Figure 6: *Rise in AI Skill Requirements in Job Postings, 2017–2025*

Notes: The left panel plots the share of postings with explicit AI requirements and the share with AI-use intensity of at least three. The right panel plots the same series as indices with 2017 equal to 100 on a log scale. Both measures rise sharply after 2017. Source: LinkUp.

The worker response exhibits a corresponding human-capital gradient. Figure 7 reports separate post-2017 difference-in-differences estimates for middle-skill workers by education. The High-AI switching differential is 0.56 percentage points for workers with a high school education or less, 0.24 percentage points for workers with some college or an associate degree, and approximately zero for workers with a bachelor's degree. These three groups account for 97 percent of the sample. Appendix Section A.10 describes the specification, and Appendix Table A.6 reports the corresponding estimates.

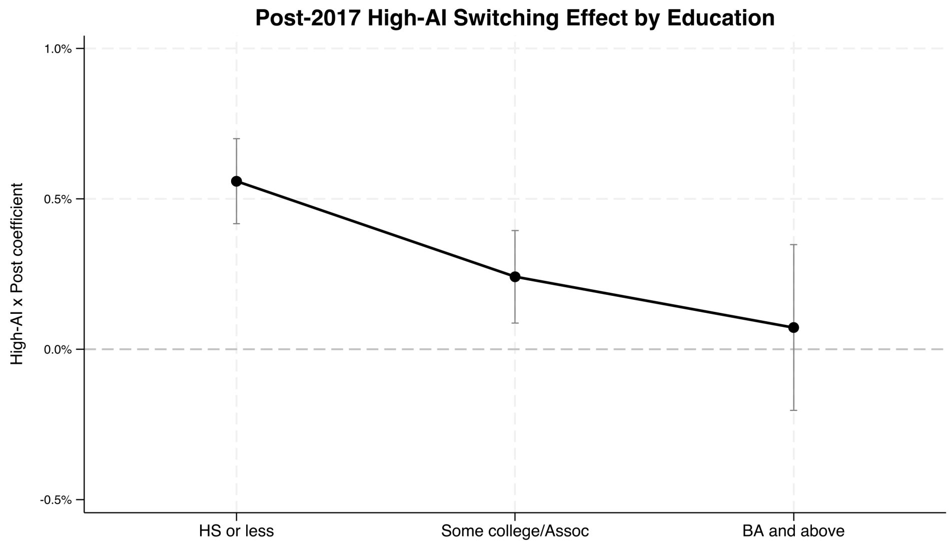


Figure 7: *Post-2017 High-AI Switching Effect by Education*
Notes: The figure reports separate difference-in-differences estimates of the post-2017 High-AI switching effect for middle-skill workers by education group. Bars denote 95 percent confidence intervals. The response declines sharply with educational attainment. Source: CPS.

Taken together, the posting and worker-level evidence is consistent with a human-capital sorting mechanism. As the AI frontier advances, AI-specific skills become increasingly important in exposed jobs. Workers with greater human capital are better positioned to learn and adapt to these technologies, while workers with less human capital are more likely to fall behind the frontier [Hassan et al., 2026, Deming and Noray, 2020, Braxton and Taska, 2023]. Consistent with this mechanism, the post-2017 switching response from high-AI middle-skill occupations declines sharply with education. The model formalizes this race between worker human capital and the AI frontier: workers who keep pace remain in or move toward High-AI employment, while workers whose skills fail to keep pace switch toward Low-AI occupations and incur earnings losses.

4 Model

I develop a frictional labor market model with High-AI and Low-AI jobs. Workers differ in human capital and employment status, firms choose which type of vacancy to create, and workers can reallocate across occupations through search and matching. High-AI employment advances the AI frontier, while faster AI progress raises the skill requirement for High-AI work. The interaction between worker skill accumulation and the moving frontier generates skill-dependent occupational reallocation.

Time is discrete, $t = 0, 1, 2, \ldots$. There is a unit mass of workers and a continuum of firms.

4.1 Environment and technology

A worker is unemployed (U), employed in a High-AI job (H), or employed in a Low-AI job (L). Let u_t, n_t^H, and n_t^L denote the corresponding population shares:

$$u_t + n_t^H + n_t^L = 1. \tag{3}$$

The High-AI frontier, $z_t^H > 0$, evolves through learning by using [Wang and Wong, 2026]. High-AI employment determines frontier growth:

$$z_{t+1}^H = (1 + g_t) z_t^H, \qquad g_t = \phi n_t^H, \qquad \phi > 0. \tag{4}$$

Low-AI technology is a fixed fraction of the High-AI frontier:

$$z_t^L = q_L z_t^H, \qquad q_L \in (0, 1). \tag{5}$$

A worker with human capital h produces $h^\alpha (z^j)^{1-\alpha}$ in sector $j \in \{H, L\}$. Define normalized skill as

$$x_t \equiv \frac{h_t}{z_t^H}. \tag{6}$$

Output in units of the High-AI frontier is therefore

$$y_H(x) = x^\alpha, \qquad y_L(x) = q_L^{1-\alpha} x^\alpha. \tag{7}$$

High-AI and Low-AI jobs also differ in their skill requirements. A Low-AI job requires

$$x \geq \bar{x}_L, \tag{8}$$

while a High-AI job requires

$$x \geq \bar{x}_H(g), \qquad \bar{x}_H(g) = \bar{x}_H^0 + \kappa g, \qquad \bar{x}_H^0 \geq \bar{x}_L, \quad \kappa > 0. \tag{9}$$

Thus, faster AI progress tightens the qualification requirement for High-AI employment.

4.2 Skill dynamics and obsolescence

Normalized skill evolves with a worker's employment state:

$$x_{t+1} = \psi_s x_t, \qquad s \in \{U, L, H\}, \qquad \psi_H > 1 \geq \psi_L > \psi_U > 0. \tag{10}$$

High-AI employment therefore allows workers to accumulate frontier-relevant skill more rapidly. In levels, $h_{t+1} = \psi_s(1 + g_t)h_t$.

Workers die with probability $\lambda_d \in (0, 1)$ at the end of each period and are replaced by newborn unemployed workers. Entry skill is drawn from a fixed distribution,

$$x_0 \sim \text{LogNormal}\left(\log x_e, \sigma_e^2\right). \tag{11}$$

A High-AI worker with current normalized skill x remains qualified for High-AI employment next period if

$$\psi_H x \geq \bar{x}_H(g_{t+1}). \tag{12}$$

A High-AI match is *obsolete* when this condition fails. Let o_t^H denote the mass of obsolete High-AI workers. Obsolescence is the key displacement margin in the model: an increase in the frontier growth rate can push marginal workers below the High-AI qualification threshold even when aggregate employment remains stable.

Timing. At the beginning of period t, workers produce and receive wages or unemployment benefits. High-AI workers then learn whether they will remain qualified at $t + 1$. Unemployed workers, Low-AI workers, and obsolete High-AI workers search. Matches are formed, existing matches separate with probability δ, and accepted job-to-job moves occur conditional on survival of the current match.

Workers then accumulate skill according to their beginning-of-period employment state. Death and entry occur at the end of the period.

4.3 Search and matching

Search is random and undirected. Firms post High-AI and Low-AI vacancies, v_t^H and v_t^L, and

$$v_t = v_t^H + v_t^L, \qquad \varphi_t = \frac{v_t^H}{v_t}. \tag{13}$$

Unemployed workers search with unit intensity, Low-AI workers search on the job with intensity χ_L, and obsolete High-AI workers search with intensity χ_H. Safe High-AI workers do not search. Effective search effort and market tightness are

$$s_t = u_t + \chi_L n_t^L + \chi_H o_t^H, \qquad \theta_t = \frac{v_t}{s_t}. \tag{14}$$

Matches form according to the constant-returns matching function

$$M(s, v) = \mu s^{\xi} v^{1-\xi}, \qquad \mu > 0, \quad \xi \in (0, 1). \tag{15}$$

The corresponding vacancy-filling and job-finding probabilities are

$$q(\theta) = \mu\theta^{-\xi}, \qquad p(\theta) = \mu\theta^{1-\xi}. \tag{16}$$

Parameters are restricted so that both probabilities lie in $[0, 1]$ over the relevant equilibrium region. Conditional on a contact, a worker meets a High-AI vacancy with probability φ_t and a Low-AI vacancy with probability $1 - \varphi_t$.

4.4 Worker values and occupational choice

Workers are risk neutral and discount future payoffs at rate β. Because workers die with probability λ_d, define

$$\tilde{\beta} \equiv \beta(1 - \lambda_d). \tag{17}$$

I impose

$$\tilde{\beta}(1 + \phi) < 1, \tag{18}$$

which guarantees finite normalized continuation values for every feasible $n_t^H \leq 1$.

Let $U_t(x)$ denote the value of unemployment, and let $W_t^j(x)$ and $J_t^j(x)$ denote the worker and firm values of a match in sector $j \in \{H, L\}$. Total match surplus is

$$S_t^j(x) = W_t^j(x) + J_t^j(x) - U_t(x). \tag{19}$$

Wages split match surplus through generalized Nash bargaining:

$$W_t^j(x) - U_t(x) = \eta S_t^j(x), \qquad J_t^j(x) = (1-\eta) S_t^j(x), \qquad \eta \in (0,1). \tag{20}$$

To characterize worker decisions on a balanced-growth path, I normalize values by the High-AI frontier z_t^H. Let $\tilde{U}(x)$ and $\tilde{S}^j(x)$ denote normalized unemployment and match-surplus values. The unemployment value satisfies

$$\tilde{U}(x) = b + \tilde{\beta}(1+g)\left[\tilde{U}(\psi_U x) + \Omega(x)\right], \tag{21}$$

where $\Omega(x)$ is the option value of search.

For an employed worker, normalized match surplus satisfies

$$\tilde{S}^j(x) = y_j(x) - b + \tilde{\beta}(1+g)\left[\mathcal{C}^j(x) - \Omega(x)\right], \qquad j \in \{H, L\}, \tag{22}$$

where $\mathcal{C}^j(x)$ collects the continuation value of the current match, including exogenous separation, occupational mobility, and the High-AI qualification constraint. Appendix B provides the explicit expressions for $\Omega(x)$ and $\mathcal{C}^j(x)$.

An unemployed worker accepts a type-j offer when the match is feasible and generates nonnegative surplus. A Low-AI worker accepts a High-AI offer when the destination job is feasible and the High-AI match yields a higher continuation value than the current Low-AI match. An obsolete High-AI worker accepts a Low-AI offer when the Low-AI qualification condition is satisfied and the new match generates nonnegative surplus.

These decisions generate four gross flows: m_t^{UH} and m_t^{UL} from unemployment into employment, m_t^{LH} from Low-AI to High-AI employment, and m_t^{HL} from obsolete High-AI to Low-AI employment.

4.5 Employment dynamics

Given the accepted worker flows, employment evolves according to

$$\begin{aligned} n^H_{t+1} &= (1-\lambda_d)\left[(1-\delta)\left(n^H_t - o^H_t + m^{LH}_t\right) + m^{UH}_t\right], \\ n^L_{t+1} &= (1-\lambda_d)\left[(1-\delta)\left(n^L_t - m^{LH}_t + m^{HL}_t\right) + m^{UL}_t\right], \\ u_{t+1} &= 1 - n^H_{t+1} - n^L_{t+1}. \end{aligned} \tag{23}$$

The cross-sectional distribution of normalized skill evolves under the same transition rules together with state-dependent skill accumulation, death, and entry.

4.6 Vacancy creation and equilibrium

Posting a type-j vacancy costs $c_j z^H_t$. Let $\mathcal{V}_j(\theta, \varphi, G)$ denote the normalized expected value of a type-j vacancy before paying the posting cost. Free entry implies

$$\mathcal{V}_j(\theta, \varphi, G) - c_j \le 0, \qquad v_j \ge 0, \qquad v_j\left[\mathcal{V}_j(\theta, \varphi, G) - c_j\right] = 0, \qquad j \in \{H, L\}. \tag{24}$$

When both vacancy types are active, the two free-entry conditions hold with equality and jointly determine labor market tightness and the vacancy composition. At a corner, the inactive vacancy type has weakly negative net posting value.

A balanced-growth equilibrium consists of employment shares, labor market tightness, the vacancy composition, the cross-sectional skill distribution, and worker and firm value functions such that worker decisions are optimal, firms satisfy free entry, the employment and skill distributions are stationary in normalized units, and

$$g^* = \phi n^{H*}. \tag{25}$$

The model contains the two forces needed to interpret the empirical results. High-AI jobs operate closer to the technological frontier and allow workers to accumulate frontier-relevant skill more rapidly, which makes them attractive to workers who satisfy the qualification requirement. At the same time, faster AI progress raises $\bar{x}_H(g)$, so marginal High-AI workers can fall behind the frontier and reallocate toward Low-AI work. Human capital therefore determines which workers move toward High-AI employment and which workers move away from it, matching the skill-dependent reallocation documented in Section 3.

5 Conclusion

This paper documents that artificial intelligence reshapes the U.S. labor market along a skill dimension that aggregate employment measures can miss. After the 2017 Transformer shock, the employment share of high-exposure occupations falls by 3.3 percentage points among middle-skill workers but rises by 3.0 percentage points among high-skill workers. Occupational switching, rather than unemployment, drives the divergence. Middle-skill workers who move away from high-exposure occupations experience earnings losses of about 7 percent relative to comparable stayers, while high-skill workers move toward high-exposure occupations without a significant earnings change. Job postings show a sharp rise in employer demand for AI-related skills, and the switching response among middle-skill workers declines steeply with education.

These facts support a human-capital sorting interpretation. I formalize this mechanism in a frictional labor market model in which High-AI employment advances the AI frontier and AI progress raises the skill requirement for High-AI work. Workers who accumulate sufficient frontier-relevant human capital remain in or move toward High-AI jobs, while marginal workers who fail to keep pace reallocate toward Low-AI work. The framework connects the worker-level adjustment documented in the data to the endogenous evolution of AI technology and provides a basis for future quantitative analysis of its longer-run distributional and growth consequences.

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A Additional Empirical Results

A.1 Occupation Groups and Sample Coverage

Table A.1: *Occupation Groups and Sample Coverage*

Group	Skill	AI Exposure	Examples	BLS-O*NET	CPS
Group 1	Low	–	Food preparation	30	27
Group 2	Middle	Low	Construction, massage therapy	194	178
Group 3	Middle	High	Administrative support, customer service	207	182
Group 4	High	Low	Art directors, social workers	112	119
Group 5	High	High	Economists, computer managers	113	127
Total				656	633

Notes: Skill requirements follow O*NET Job Zones. Job Zone 1 defines low-skill occupations, Job Zones 2–3 define middl skill occupations, and Job Zones 4–5 define high-skill occupations. AI exposure follows the occupation-level SML measu and uses the within-skill median as the cutoff between low and high exposure. BLS-O*NET counts refer to detailed SC six digit occupations. CPS occupations are mapped to SOC codes with the official Census crosswalk.

Table A.2: *Employment Dynamics by Group: BLS versus CPS*

	Level (Millions)		Share (%)	
Group	2017	2024	2017	2024
Panel A: BLS (OEWS)				
Group 1 (JZ 1 All)	5.34	5.49	4.73	4.60
Group 2 (JZ 2–3 Low AI)	33.34	34.83	29.52	29.20
Group 3 (JZ 2–3 High AI)	43.56	42.06	38.57	35.27
Group 4 (JZ 4–5 Low AI)	9.55	11.01	8.46	9.23
Group 5 (JZ 4–5 High AI)	21.14	25.89	18.72	21.70
Panel B: CPS				
Group 1 (JZ 1 All)	6.59	5.28	5.70	4.39
Group 2 (JZ 2–3 Low AI)	28.89	30.61	24.97	25.41
Group 3 (JZ 2–3 High AI)	34.21	31.04	29.57	25.77
Group 4 (JZ 4–5 Low AI)	17.04	19.18	14.73	15.92
Group 5 (JZ 4–5 High AI)	28.96	34.34	25.03	28.51

Notes: The table compares employment dynamics across two data sources. Panel A uses BLS OEWS estimates (2017–2024). Panel B uses CPS microdata (2017–2024). Both frames show the fall in the Group 3 share and the rise in the Group 5 share.

A.2 Validation of Exposure and Skill Measures

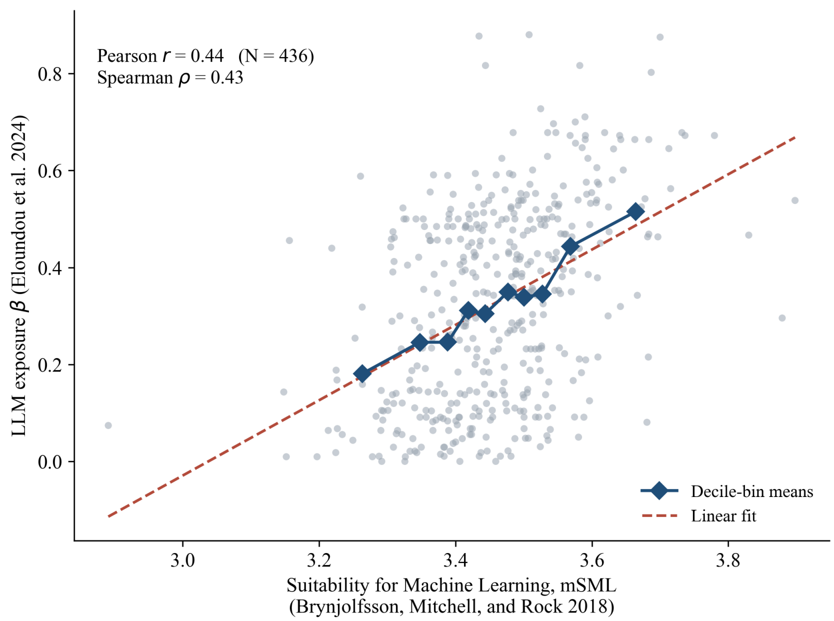


Figure A.1: *Validation of the SML Exposure Measure Against LLM Exposure*

Notes: Each gray dot is one of the 436 six-digit SOC occupations covered by both measures. The horizontal axis is the occupation-level suitability for machine learning score (mSML) of Brynjolfsson et al. [2018]. The vertical axis is the occupation-level LLM exposure share β of Eloundou et al. [2023]. Diamonds show the mean of β within each decile of the mSML distribution, and the dashed line is the OLS fit. The unweighted correlation is 0.44 (Spearman 0.43; 0.64 when weighted by 2024 OEWS employment), and mean LLM exposure rises monotonically across mSML deciles. The positive relationship arises because both measures ultimately score the same underlying task content of occupations: the prevalence of structured, information-processing tasks that machine learning can perform.

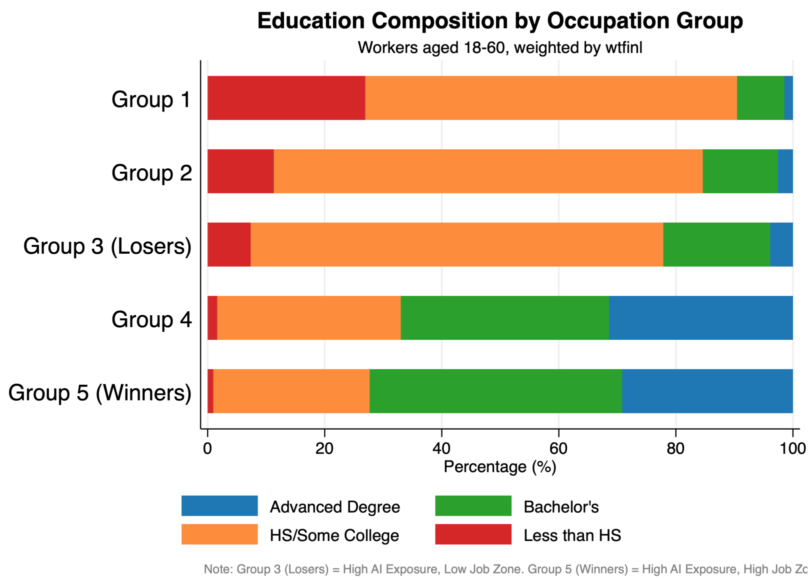


Figure A.2: *Education Level by Occupation Group*
Notes: The figure reports the education composition of workers aged 18 to 60 in each occupation group, with CPS final weights (`wtfinl`). Groups 4 and 5 consist predominantly of workers with a bachelor's or advanced degree; Groups 2 and 3 consist predominantly of workers with a high school diploma or some college. The composition is nearly identical across exposure levels within a skill tier. Source: CPS.

A.3 Timing of the AI Shock

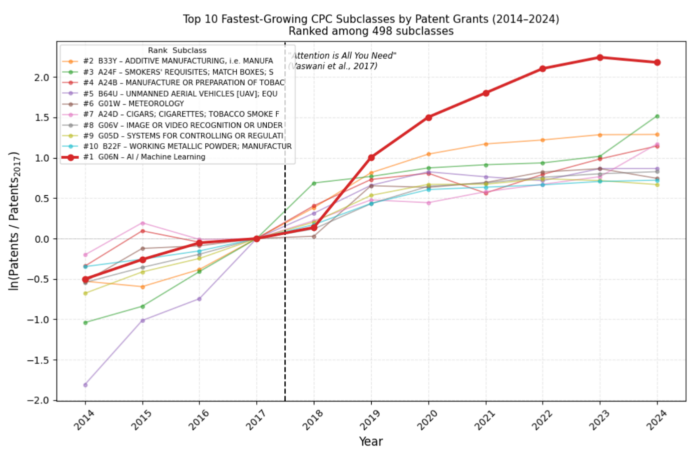


Figure A.3: *Patent Growth around the 2017 Transformer Shock*
Notes: The figure plots patent growth for the fastest-growing Cooperative Patent Classification subclasses. The AI and machine-learning subclass, G06N, exhibits the strongest post-2017 acceleration. Source: USPTO.

A.4 Additional Employment Dynamics

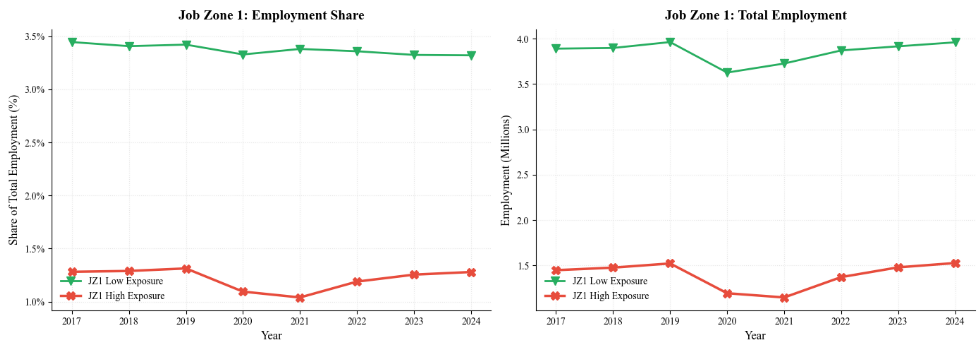


Figure A.4: *Employment Dynamics for Job Zone 1*

Notes: The figure plots employment shares and levels for Job Zone 1 occupations by exposure level. Both series are flat over the sample, apart from the pandemic dip, which motivates the exclusion of Group 1 from the main two-by-two analysis. Source: BLS OEWS.

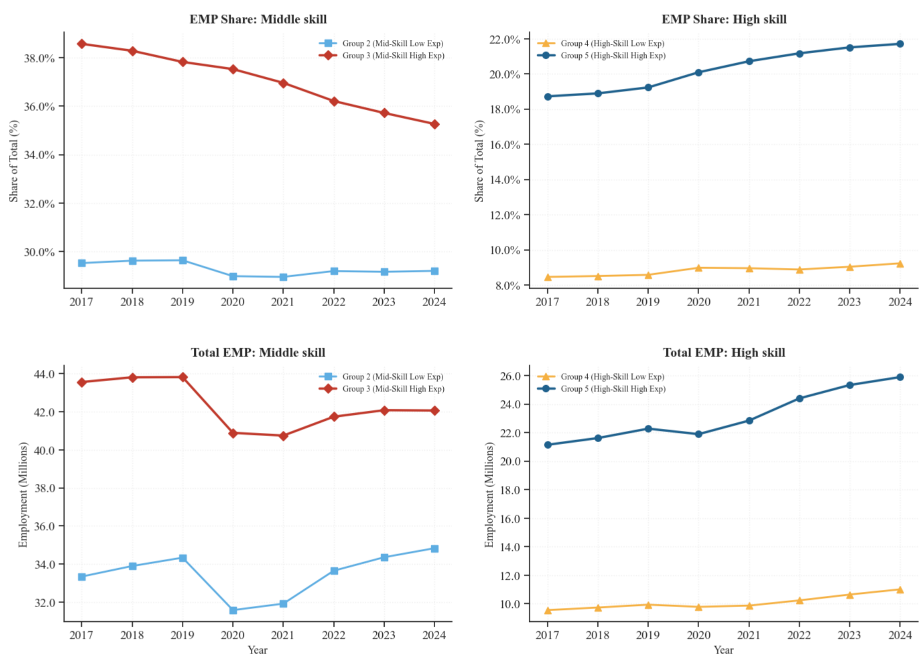


Figure A.5: *Employment Share and Level for the Two Skill Tiers: BLS*

Notes: The figure plots employment shares and levels by group under the BLS OEWS frame. Source: BLS OEWS.

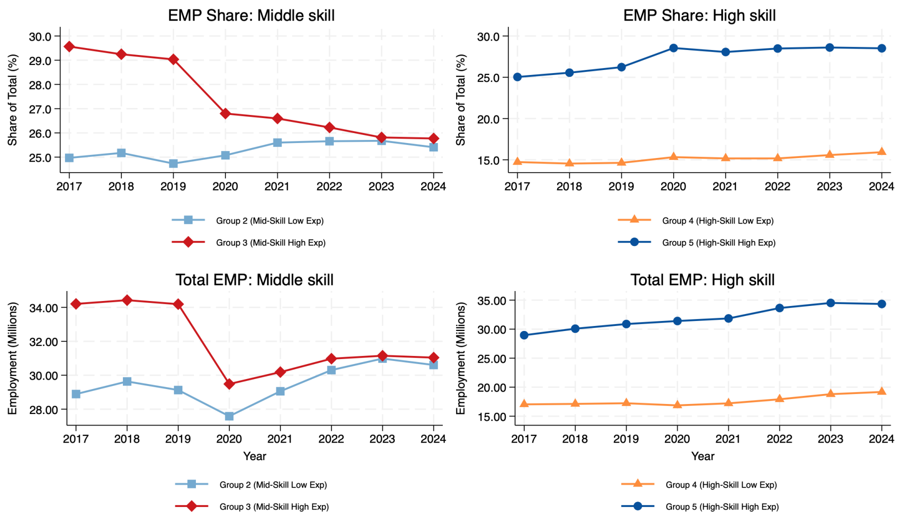


Figure A.6: *Employment Share and Level for the Two Skill Tiers: CPS*
Notes: The figure replicates the group-level dynamics with CPS data. The CPS reproduces the fall in the Group 3 share and the rise in the Group 5 share. Source: CPS.

Table A.3: *Shift in Employment Shares within Skill Group*

Year	Share (%)			Δ from 2017 (pp)		
	Low AI	High AI	NIE	Δ Low AI	Δ High AI	Δ NIE
Panel A: Mid-Skill Workers (HS to Associate's Degree)						
2017	43.44	50.96	5.60	0.00	0.00	0.00
2018	44.09	50.80	5.10	+0.65	−0.15	−0.50
2019	43.93	51.25	4.81	+0.49	+0.30	−0.79
2020	43.18	45.78	11.03	−0.26	−5.17	+5.43
2021	45.63	47.14	7.22	+2.19	−3.81	+1.62
2022	46.99	47.87	5.13	+3.55	−3.08	−0.47
2023	47.48	47.57	4.95	+4.03	−3.38	−0.65
2024	47.12	47.57	5.31	+3.68	−3.39	−0.29
Panel B: High-Skill Workers (Bachelor's Degree or Higher)						
2017	36.48	60.95	2.57	0.00	0.00	0.00
2018	35.71	61.80	2.48	−0.76	+0.85	−0.08
2019	35.27	62.30	2.44	−1.21	+1.34	−0.13
2020	33.29	61.25	5.47	−3.19	+0.30	+2.90
2021	34.20	62.47	3.33	−2.28	+1.52	+0.76
2022	34.22	63.35	2.43	−2.26	+2.39	−0.14
2023	34.72	62.82	2.46	−1.76	+1.87	−0.10
2024	35.13	62.14	2.73	−1.35	+1.19	+0.17

Notes: Shares are calculated within each skill pool (mid-skill or high-skill), weighted by CPS final weights (`wtfinl`). "Low AI" and "High AI" refer to employed workers in low and high AI-exposure occupations, respectively. "NIE" (not in employment) includes unemployed and not-in-labor-force individuals, classified by education level. Mid-skill: high school diploma to associate's degree; high-skill: bachelor's degree or higher. The sample consists of prime-age individuals (18–60). The 2020 movements reflect the pandemic. Source: CPS.

A.5 LinkUp Postings: Representativeness and Posting Shares

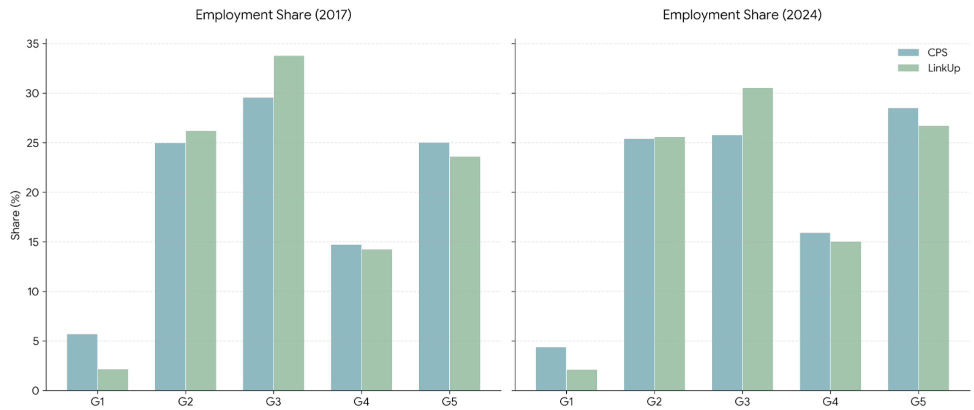
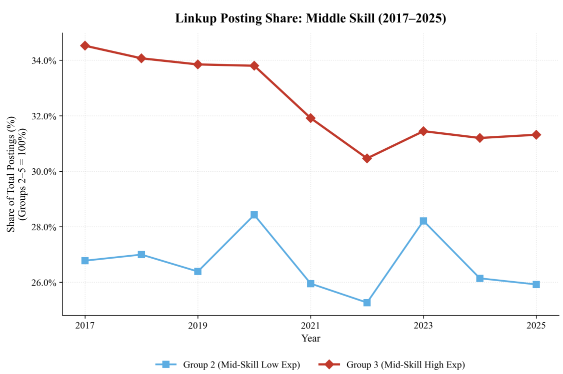
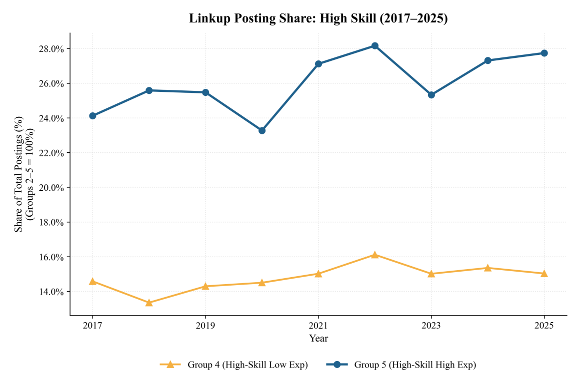


Figure A.7: *Representativeness of the LinkUp Sample*
Notes: The figure compares group shares in the CPS and in LinkUp postings for 2017 and 2024. The group structure of the posting sample matches the household survey in both years.



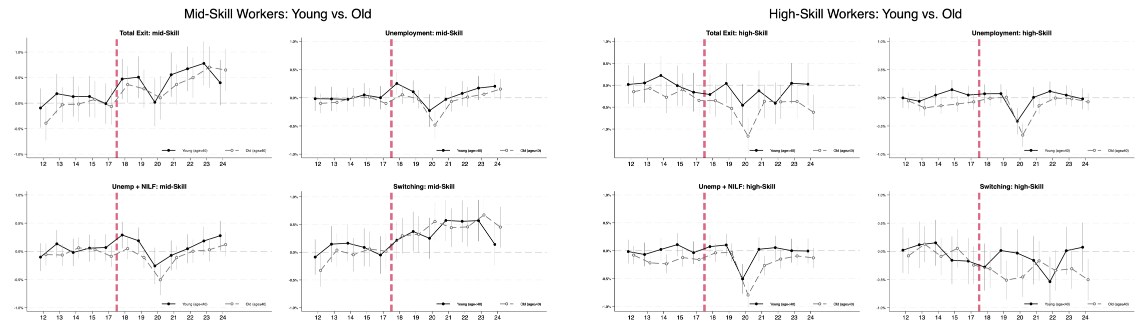
(a) *Middle-skill workers*



(b) *High-skill workers*

Figure A.8: *Posting Shares by Group in LinkUp: The Demand Side*
Notes: The figure plots the share of LinkUp postings by occupation group, with Groups 2 through 5 normalized to 100 percent. The posting share of Group 3 falls and the posting share of Group 5 rises over 2017 to 2025, in parallel with the employment shares in Figure 1. Source: LinkUp.

A.6 Event Studies by Age



(a) *Middle skill: young versus old*

(b) *High skill: young versus old*

Figure A.9: *Event Studies by Age: Young versus Old Workers*

Notes: The figure repeats the event studies of Figures 3 and 4 for workers below and above age 40. Switching among exposed middle-skill workers is concentrated among older workers. Among high-skill workers, in contrast, the relative movement toward high-AI occupations is stronger among younger workers. Source: CPS.

A.7 Inference Robustness: Occupation-Level Clustering

The baseline specifications cluster standard errors at the individual level, which accounts for serial correlation within workers across survey months. Because AI exposure is assigned at the occupation level, however, workers in the same occupation share common shocks, and the conservative choice is to cluster at the level of treatment assignment [Abadie et al., 2023]. Table A.4 re-estimates the main difference-in-differences specifications with standard errors clustered by CPS Census occupation code (388 clusters in the middle-skill sample, 259 in the high-skill sample, pooling the 2010 and 2018 Census coding schemes). Point estimates are identical by construction; standard errors roughly double. All four headline coefficients, the increases in switching and total exit for middle-skill exposed workers and the declines for high-skill exposed workers, remain significant at the 5 percent level or better. The unemployment and nonparticipation margins remain economically and statistically negligible.

Table A.4: *Main Estimates with Occupation-Level Clustering*

	Total Exit	Unemployment	Unemp. + NILF	Switching
Panel A: Middle-skill (Group 3 vs. Group 2)				
High-AI × Post	0.38^{***}	0.03	0.01	0.34^{***}
clustered by individual	(0.06)	(0.03)	(0.03)	(0.05)
clustered by occupation	(0.13)	(0.08)	(0.09)	(0.10)
Panel B: High-skill (Group 5 vs. Group 4)				
High-AI × Post	-0.31^{**}	-0.02	-0.06	-0.27^{**}
clustered by individual	(0.06)	(0.02)	(0.03)	(0.06)
clustered by occupation	(0.12)	(0.05)	(0.05)	(0.12)

Notes: Coefficients and standard errors (×100) are in percentage points of monthly transition probability. Specifications are identical to the baseline tables; only the clustering of standard errors varies. Occupation clustering uses CPS Census occupation codes, pooling the 2010 and 2018 coding schemes (388 clusters in Panel A, 259 in Panel B). Stars refer to the occupation-clustered standard errors. Observations: 2,652,691 (Panel A) and 2,153,323 (Panel B). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: CPS.

A.8 IPW Robustness: Clustering and Trimming

Table A.5 subjects the wage-consequence estimates to the same clustering change and to a stricter trimming rule. Column (1) repeats the baseline estimate (propensity scores trimmed at the 2nd and 98th percentiles, standard errors clustered by individual). Column (2) clusters by origin occupation, column (3) trims at the 1st and 99th percentiles instead, and column (4) combines both. The earnings loss from moves out of middle-skill high-exposure work into low-exposure work (Group 3 to Group 2) is stable at roughly −\$1,970 across all four variants and remains significant at the 5 percent level; the estimate for upward moves (Group 4 to Group 5) remains small and insignificant throughout.

Table A.5: *IPW Wage Estimates: Clustering and Trimming Robustness*

	(1) Baseline	(2) Occ. cluster	(3) Trim 1/99	(4) Trim 1/99, occ. cluster
G3 $\rightarrow$ G2	-1.971^{**}	-1.971^{**}	-1.976^{**}	-1.976^{**}
	(0.881)	(0.797)	(0.868)	(0.786)
Observations	23,858	23,858	24,356	24,356
G4 $\rightarrow$ G5	-0.696	-0.688	-0.918	-0.918
	(1.365)	(1.353)	(1.347)	(1.312)
Observations	14,783	14,783	15,091	15,091

Notes: Each cell reports the coefficient on the mover indicator from the full IPW specification (demographic controls and industry-by-year fixed effects), in thousands of 2023 dollars of real annual earnings change. Column (1) is the baseline: propensity scores trimmed at the 2nd and 98th percentiles, standard errors clustered by individual. Column (2) clusters standard errors by the worker's origin occupation. Column (3) trims propensity scores at the 1st and 99th percentiles. Column (4) applies both changes. Propensity scores, weights, and controls are re-estimated within each variant. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: CPS.

A.9 External Evidence on Observed AI Use

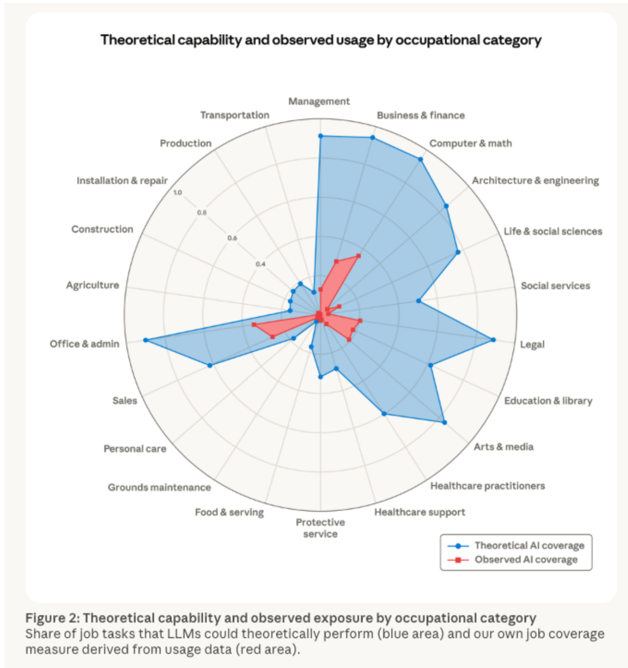


Figure A.10: *Observed AI Use versus Theoretical Capability by Occupation*
Notes: The figure reproduces the share of job tasks that large language models could in theory perform against a job coverage measure from observed usage data, by occupational category. Observed use concentrates in computer, mathematical, business, and legal occupations. Data source: Anthropic report, March 5, 2026.

A.10 Human-Capital Heterogeneity in Worker Reallocation

Education-gradient specification. Figure 7 reports separate difference-in-differences estimates by education group within the middle-skill sample. For each education group g, I estimate

$$Y_{it} = \alpha + \beta_g \left(\text{HighAI}_{it} \times \text{Post}_t\right) + \gamma\, \text{HighAI}_{it} + X'_{it}\delta + \lambda_y + \mu_m + \phi_j + \varepsilon_{it}, \quad \text{(A.1)}$$

where Y_{it} is an indicator for an occupational switch between consecutive months. The sample contains employed middle-skill workers in Groups 2 and 3. The controls, fixed effects, and clustering follow the baseline transition specification. The coefficient β_g measures the post-2017 change in switching from high- relative to Low-AI occupations within education group g.

Table A.6: *Post-2017 High-Exposure Switching Effect by Education*

	(1) HS or less	(2) Some college/Assoc.	(3) BA and above
High-AI × Post	0.559***	0.241***	0.072
	(0.072)	(0.079)	(0.141)
Baseline controls	Yes	Yes	Yes
Year, month, industry FE	Yes	Yes	Yes
Observations	1,221,870	956,215	474,606

Notes: Each column reports the coefficient on High-AI × Post from Equation (A.1), estimated separately by education group among middle-skill workers in Groups 2 and 3. Coefficients and standard errors are expressed in percentage points of monthly switching probability. The three estimates correspond to the coefficients plotted in Figure 7. All specifications include the baseline worker controls and year, month, and industry fixed effects. Standard errors, clustered at the individual level, appear in parentheses. Source: CPS. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Model Value Functions

This appendix provides the explicit normalized Bellman equations that underlie the worker-choice rules in Section 4.4. All values are expressed in units of the High-AI frontier.

Define

$$I^H(x) = \mathbf{1}\{\psi_H x \geq \bar{x}_H(g)\}, \qquad \Delta^j(x) = \tilde{U}(\psi_j x) - \tilde{U}(\psi_U x), \qquad j \in \{H, L\}. \tag{B.1}$$

The indicator $I^H(x)$ equals one when a current High-AI worker remains qualified for High-AI employment next period.

For an unemployed worker, the option value of search is

$$\begin{aligned}\Omega(x) = p(\theta)\eta\Big[&\varphi\,\mathbf{1}\{\psi_U x \geq \bar{x}_H(g)\}\max\{\tilde{S}^H(\psi_U x), 0\} \\ &+ (1-\varphi)\,\mathbf{1}\{\psi_U x \geq \bar{x}_L\}\max\{\tilde{S}^L(\psi_U x), 0\}\Big].\end{aligned} \tag{B.2}$$

For an obsolete High-AI worker, the Low-AI offer rate is

$$\rho_{HL}(x) = \chi_H p(\theta)(1-\varphi)\mathbf{1}\{\psi_H x \geq \bar{x}_L\}. \tag{B.3}$$

For a Low-AI worker, define the effective poaching rate into High-AI employment as

$$m(x) = \chi_L p(\theta)\varphi\,\mathbf{1}\{\psi_L x \geq \bar{x}_H(g)\}\mathbf{1}\{\tilde{S}^H(\psi_L x) \geq \tilde{S}^L(\psi_L x)\}. \tag{B.4}$$

The normalized High-AI surplus satisfies

$$\begin{aligned}\tilde{S}^H(x) = x^\alpha - b + \tilde{\beta}(1+g)\Big[\Delta^H(x) + (1-\delta)\big(I^H(x)\tilde{S}^H(\psi_H x) \\ + [1 - I^H(x)]\rho_{HL}(x)\,\eta\max\{\tilde{S}^L(\psi_H x), 0\}\big) - \Omega(x)\Big].\end{aligned} \tag{B.5}$$

The normalized Low-AI surplus satisfies

$$\begin{aligned}\tilde{S}^L(x) = q_L^{1-\alpha}x^\alpha - b + \tilde{\beta}(1+g)\Big[\Delta^L(x) + (1-\delta)\big(m(x)\eta\tilde{S}^H(\psi_L x) \\ + [1 - m(x)]\tilde{S}^L(\psi_L x)\big) - \Omega(x)\Big].\end{aligned} \tag{B.6}$$

Equations (21), (B.5), and (B.6), together with the acceptance rules in Section 4.4, determine worker values and occupational choices on a balanced-growth path.